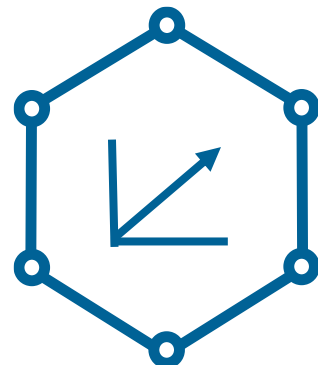


Knowledge Graphs

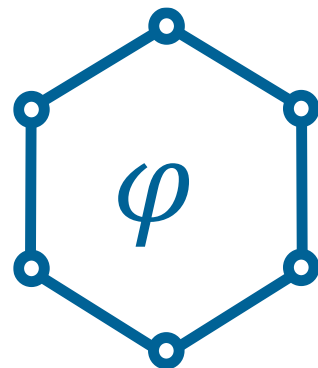


Emanuel Sallinger



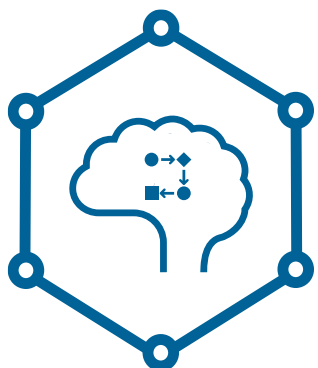
KG Embeddings

Widely-applied, large family of ML models



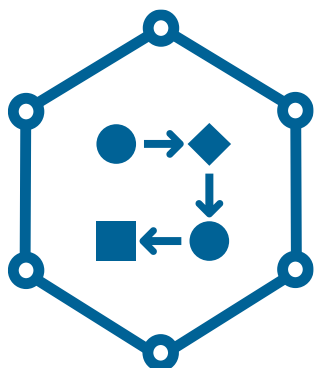
Logical Knowledge in KGs

Highly expressive, diverse family of logical models.



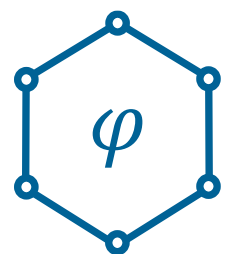
Graph Neural Networks

Using the KG structure *as* a neural network.

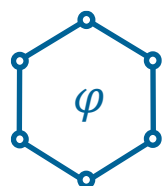


Data Models

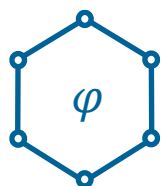
Overview of data models in different communities



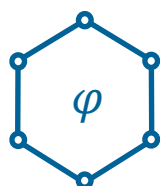
Logical Knowledge in KGs



Challenges



Recursion in **SQL**

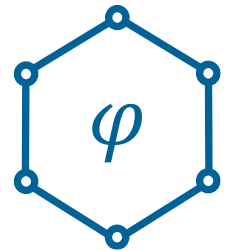


Recursion in **Datalog**

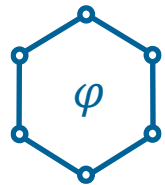


Recursion in the **Real World**

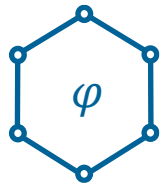




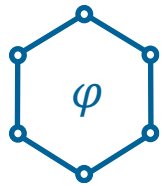
Logical Knowledge in KGs



Creation



Recursion and Creation
A Unified Framework



Recursion and Creation
Warded



The **Real World**



Theory to **Practice**